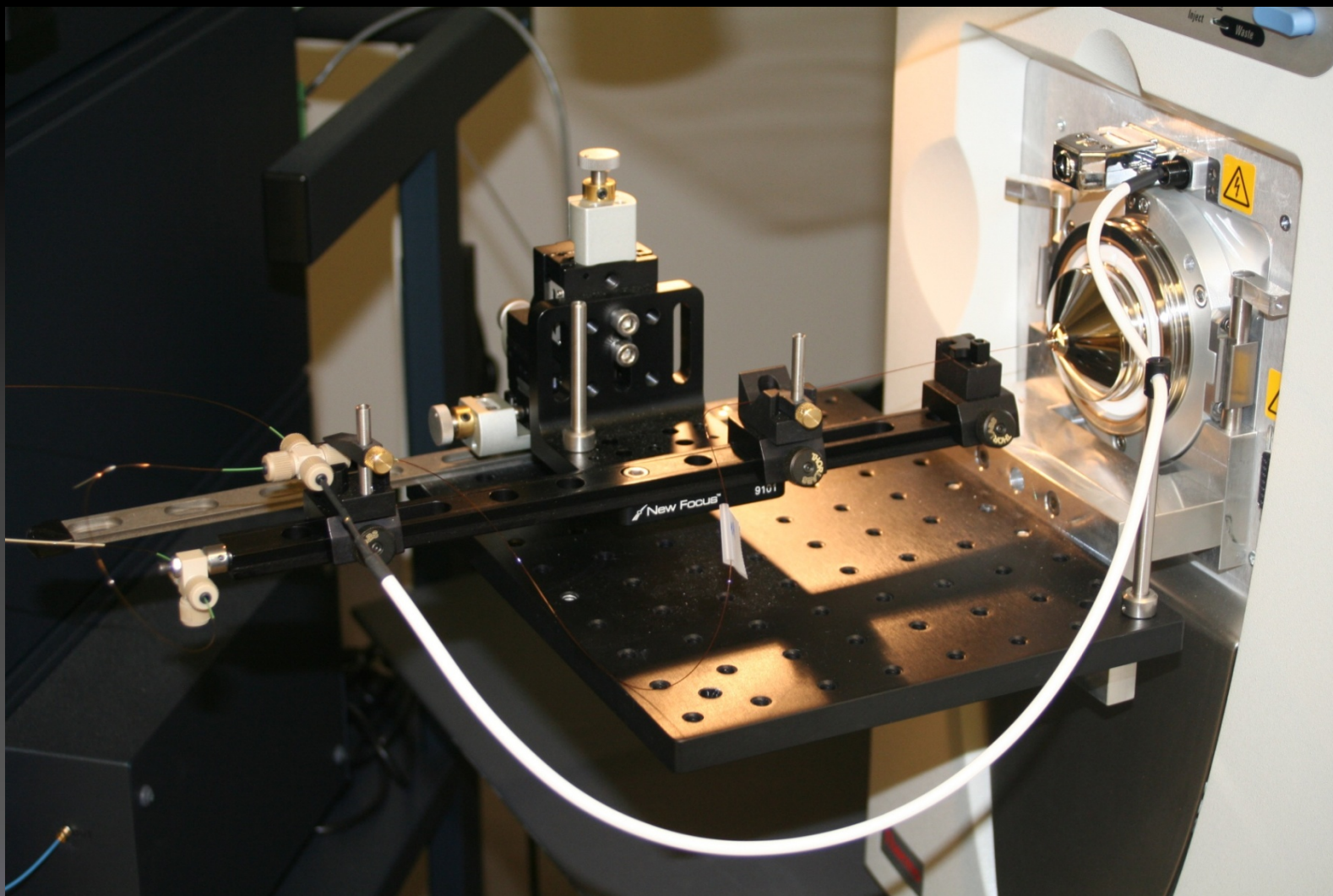


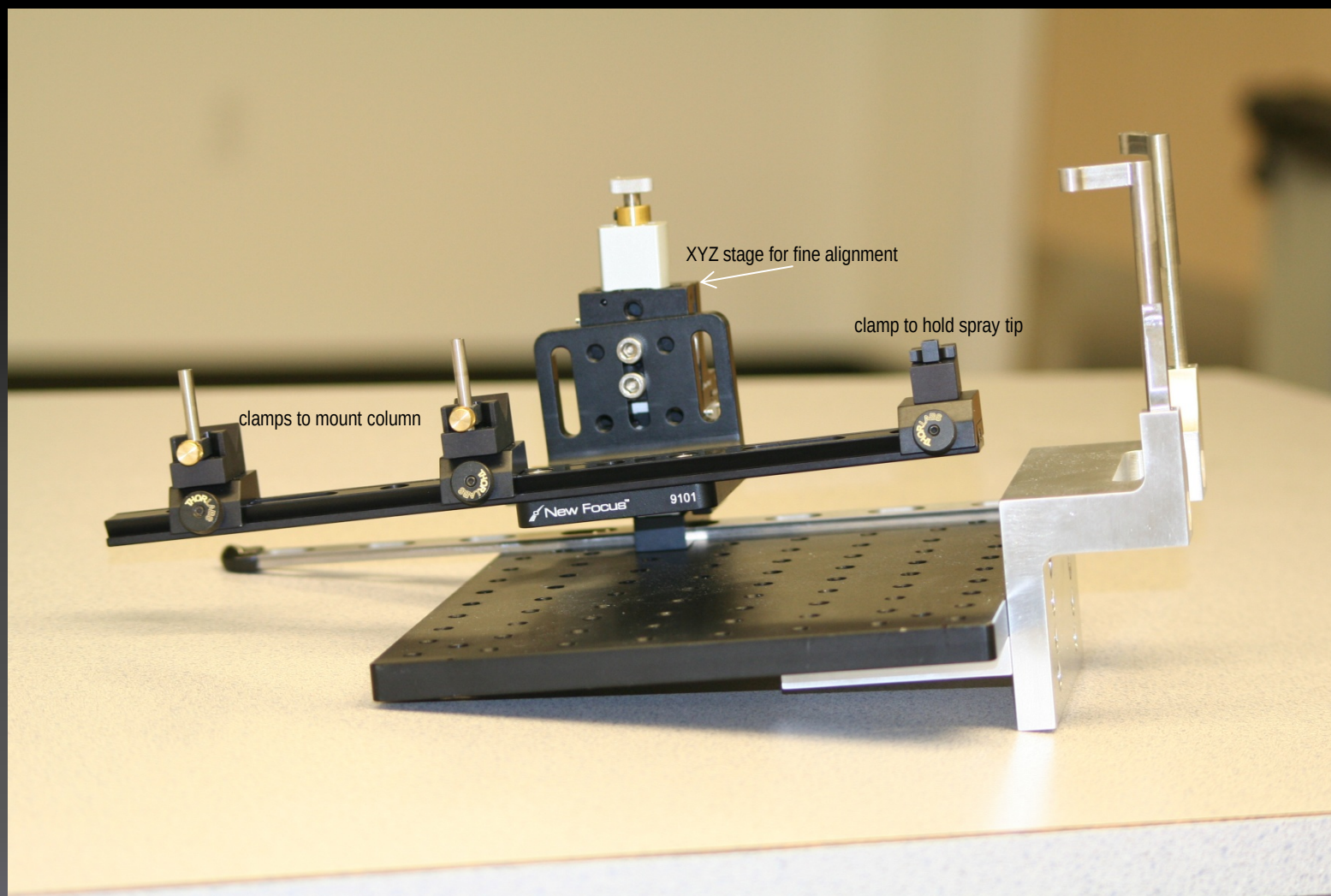
New NSI source



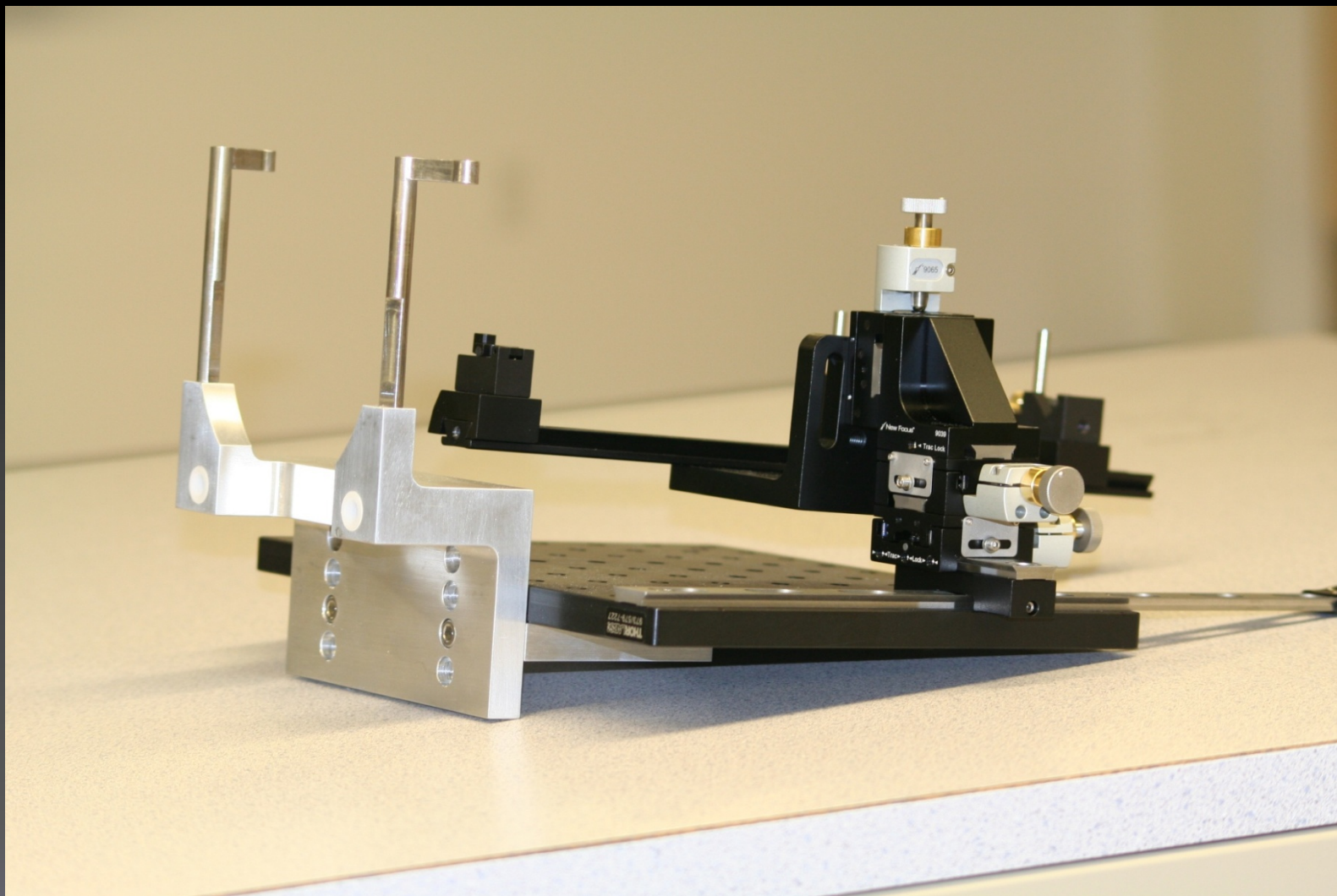
New NSI source

- The NSI source was designed to be used on the Ion Max™ source on ThermoFisher instruments (LTQ and TSQ series)
- It offers all the benefits and features of commercial NSI sources:
 - Lower flow rates: no drying gas or thermal heating is required and increased sensitivity compared to higher flow ESI
 - Higher tolerance to a wide variety of liquid compositions than conventional ESI
- Additionally this new NSI source offers significant improvements over most commercial NSI sources:
 - Great flexibility: it accommodates in house packed columns of all length, commercial columns, as well as configurations with traps, spray tips etc.
 - The source can be customized to your specific needs: e.g. mounting of cameras, LED lights or anything else you can think of.....
 - Additionally this NSI source can be used in conjunction with the Thermo NSI probes (static , dynamic and packed tip probe)
 - The open design allows for leaks to be detected immediately
 - Significantly lower cost
 - The NSI platform can also be used with the adaptor ring of the Thermo NSI source, eliminating the need to machine the adaptor bracket
 - All the parts are commercially available, except for the adaptor system (custom machined, e.g. by emachineshop.com)
- The UWPR is currently using the NSI source on all of our Thermo instruments
- The plans for this NSI source are available for free for academic and non-commercial use only

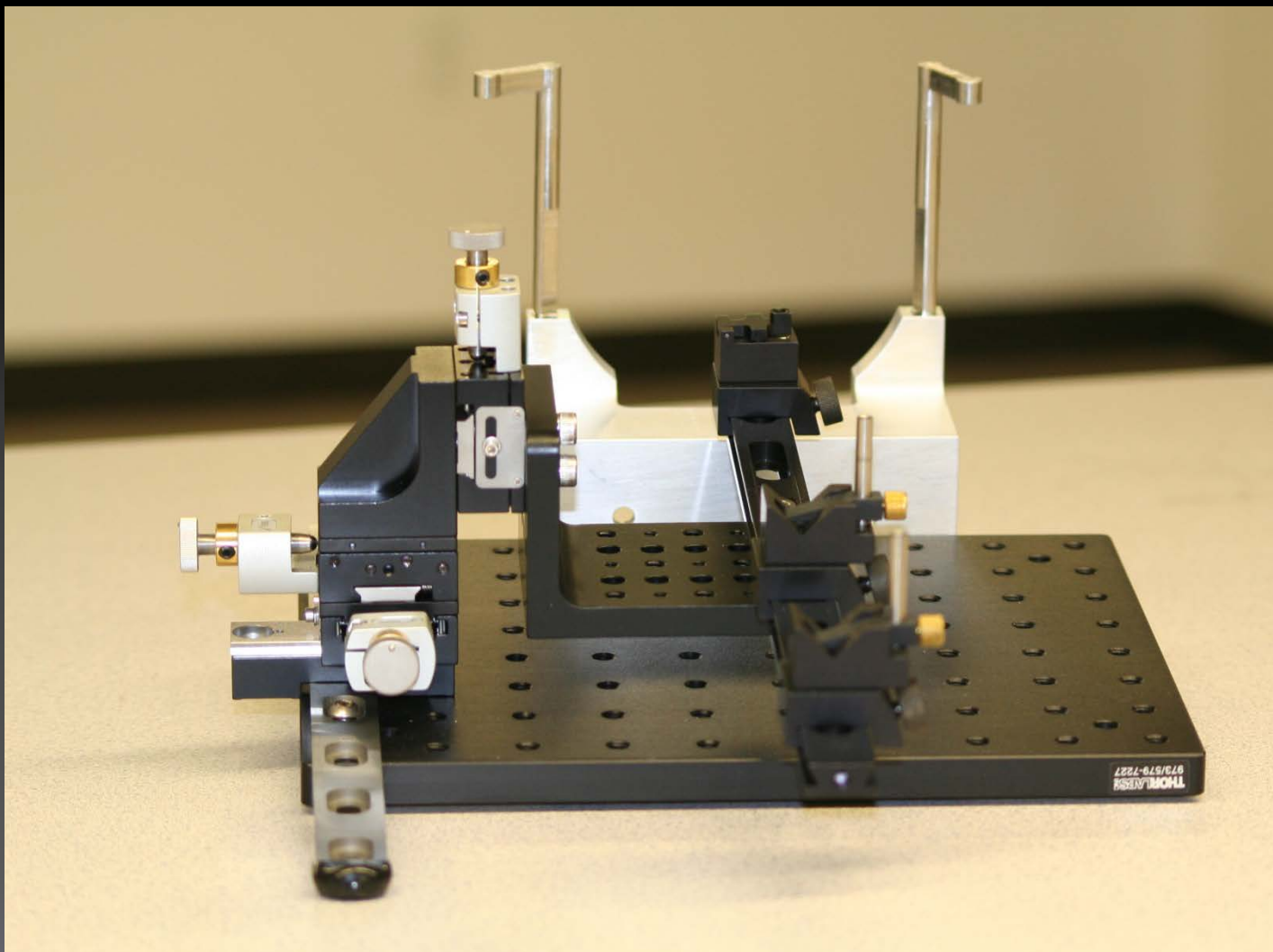
NSI source side view



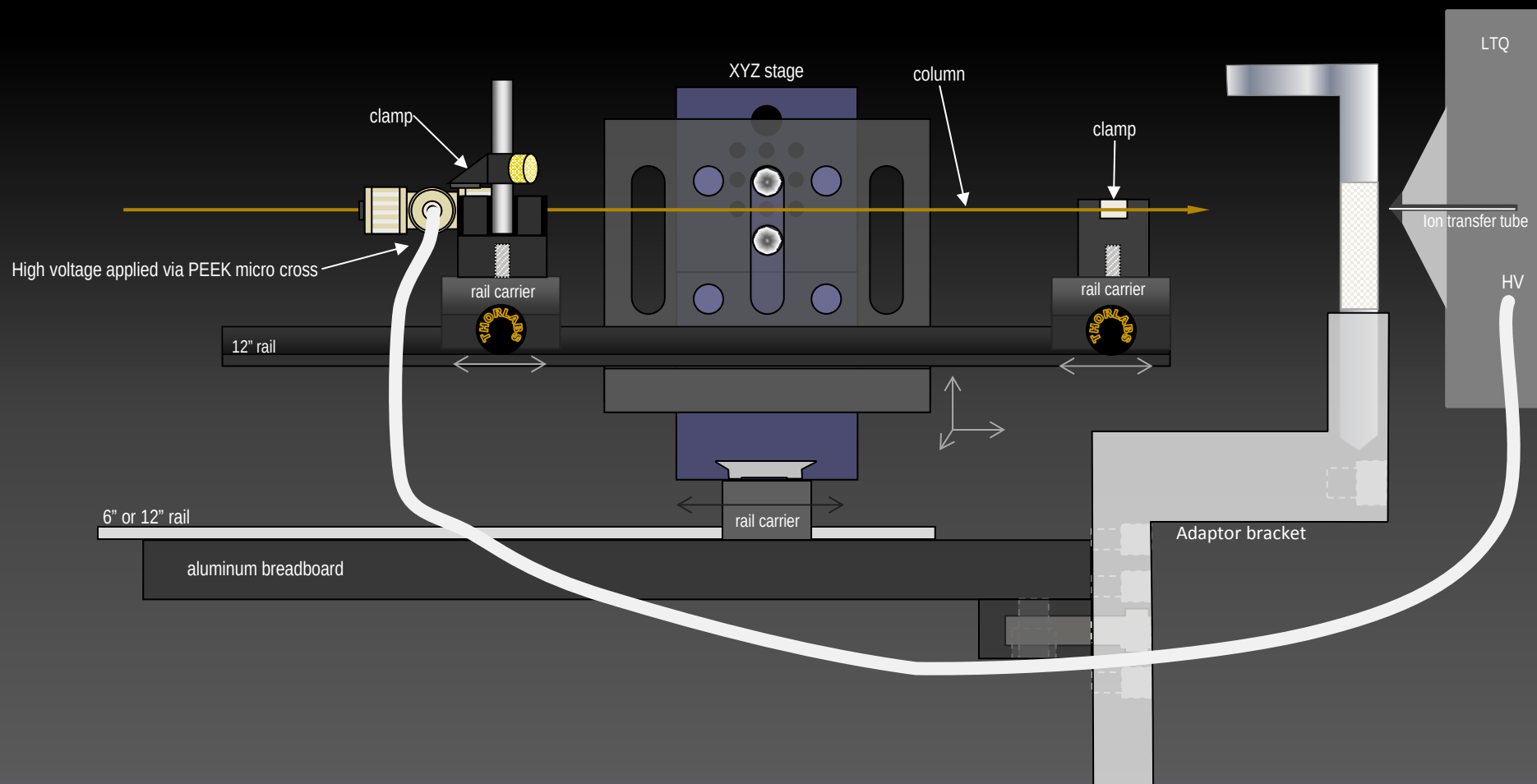
NSI source side view



NSI source back side view



NSI source schematic



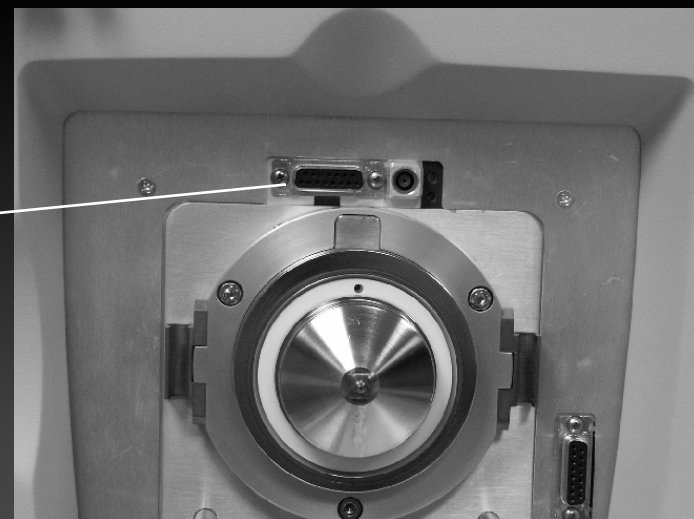
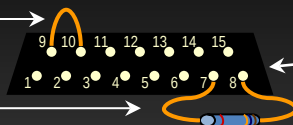
Interlock bypass

To bypass interlock on ThermoFisher LTQ when using alternate nanospray source (NSI)

Note: Jumper and resistor can either be attached directly to the connector on the front of the LTQ or via a DB15 male connector (easier to change between different sources).

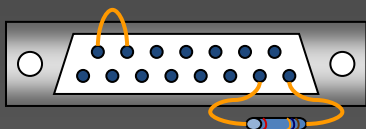
Jumper to bypass interlock between socket 9 and 10

10 K ohm resistor (2% var.) for NSI source recognition between socket 7 and 8



OR use DB15 male connector:

solder the jumper and resistor to the corresponding pins



DB15 male connector backside view



High voltage connection

IMPORTANT: cover exposed metal with heat shrink polyolefin tubing

Socket gold flash
over nickel

Pin gold flash
over nickel

0.5mm gold
or platinum wire

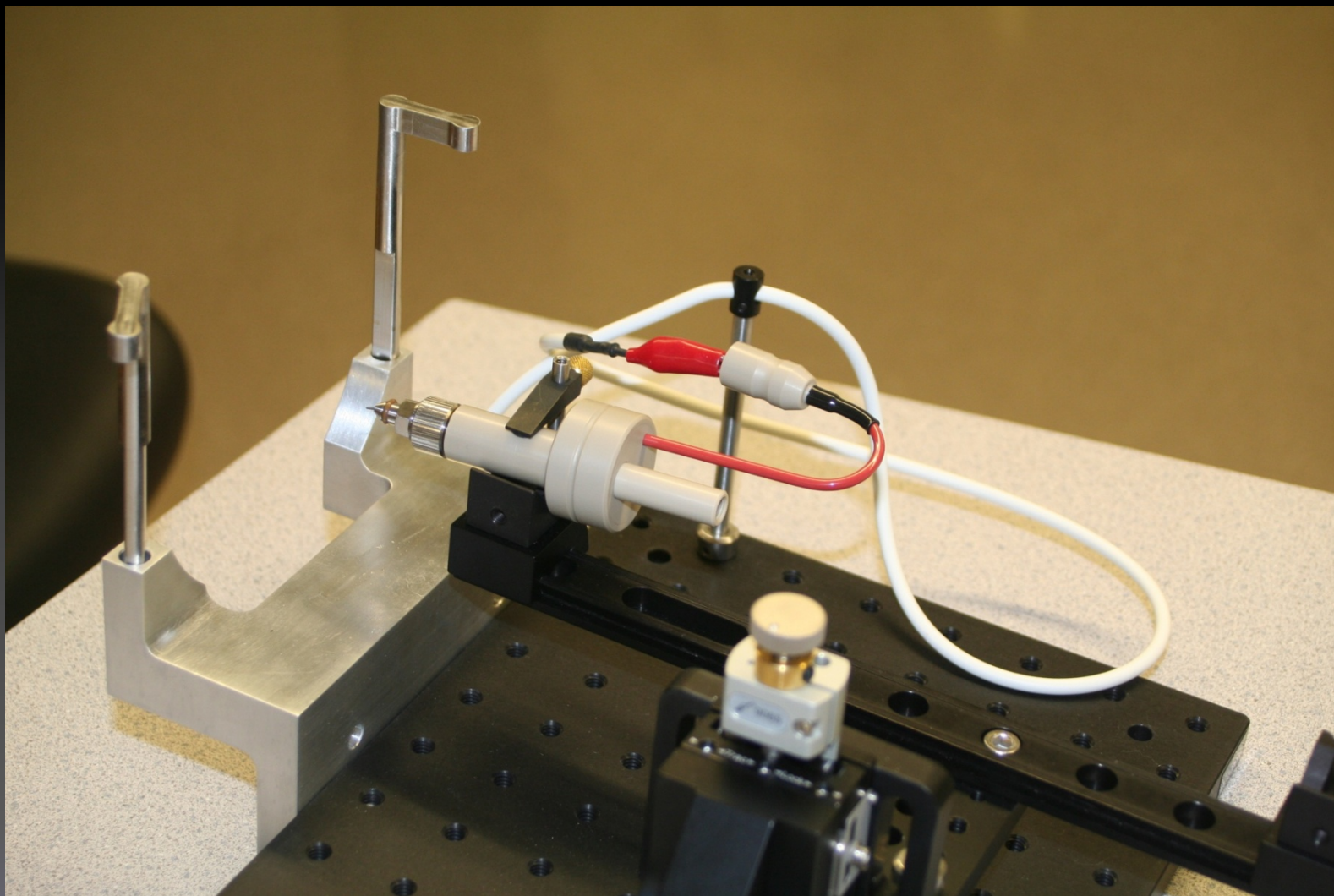
To LTQ
SCID series, threaded
plastic flange receptacle

Use Upchurch micro cross or tee
to connect HV to liquid path

High voltage wire:
#22 AWG, .180 O.D. silicone rubber
insulated wire OR
20 AWG; 19/32; 15 kV; 0.037 in.; 0.120 in.;
200 degC; Silver Plated Copper



Static Nanospray Probe



Dynamic Nanospray Adaptor

